Supplement to "Identifying Rational Types in Unknown Environments"

Appendix B: Formal Proofs and Lean Verification

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This supplement presents in full detail the formal statements and proofs supporting the identification argument in the main paper. It is organized in seven sections that mirror the logical structure of the model: public information and Bayesian learning (Section S1), the stochastic implementation mechanism (Section S2), asymptotic type identification (Section S3), competitive risks and operational pressure (Section S4), policy implications (Section S5), equilibrium implementability (Section S6), and the abstract Kakutani identification criterion (Section S7), plus machine-checked verification (Section S8). Analytical proofs are self-contained in this document; algebraic steps verified in Lean 4 are reproduced from the `appendix_B_proofs/` replication package (Section S9), available in the public repository at https://github.com/Donzhumber/TPV.git. Classical measure-theoretic inputs (Kakutani, Blackwell–Dubins, fixed-point existence) remain proved here and are not re-derived inside the proof assistant.

S1 Public information, likelihood, and Bayesian learning

The three definitions below construct the public filtration, the likelihood kernel, and the belief process. The stability lemma closes the block.

Definition S1.1 (Public signal and filtration)**. (i) Public signal ζ_t.** The public signal ζ_t is the random vector that summarizes what is observable at the close of period t:

$$\zeta_t = (\alpha_t, \gamma_t, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, m_t, d_t) \in \mathcal{Z}_t \subseteq \mathbb{R}^2 \times \mathcal{A}^F \times \mathcal{A}^K \times \mathcal{A}^{S,\text{disc}} \times \mathcal{M}_{\text{out}} \times \{0,1\}.$$

(ii) Public filtration $\{\mathcal{F}_t^{\text{pub}}\}$. Set $\mathcal{F}_0^{\text{pub}} := \sigma(z, \theta_V)$. For $t \geq 1$,

$$\mathcal{F}_t^{\text{pub}} := \sigma(z, \zeta_0, \ldots, \zeta_{t-1}),$$

that is, the σ-algebra generated by the initial location and by the signals observed strictly before period t begins.

(iii) Measurability. Each $\zeta_t : (\Omega, \mathcal{F}) \to (\mathcal{Z}_t, \mathcal{B}(\mathcal{Z}_t))$ is measurable, and $\sigma(z, \zeta_0, \ldots, \zeta_{t-1})$ is well defined as a sub-σ-algebra of $\mathcal{F}$.

□

Definition S1.2 (Predictive likelihood under a counterfactual type)**.** Fixing a PBE with profile (χ, s^F, s^K) and a counterfactual type $\theta \in \Theta_K$, the **predictive likelihood** of period t is

$$\ell_t(\theta; J) := \mathbb{P}_{\text{C},\theta}\Big(\zeta_t \in J \mid \mathcal{F}_t^{\text{pub}}\Big), \qquad J \in \mathcal{B}(\mathcal{Z}_t). \tag{S1.1}$$

The mapping $J \mapsto \ell_t(\theta; J)$ is a **probability kernel** that carries the information state $\mathcal{F}_t^{\text{pub}}$ to the law of ζ_t under type θ.

□

Definition S1.3 (Beliefs and PBE-consistent Bayesian updating)**.** Let $\mu_0 \in \Delta(\Theta_K)$ be the **common prior** with full support, $\mu_0(\theta \mid z, \theta_V) > 0$ for all $\theta \in \Theta_K$. For $t \geq 1$, let $\mu_t \in \Delta(\Theta_K)$ be the distribution obtained by iterated Bayes from μ_0, using the realizations of $(\zeta_0, \ldots, \zeta_{t-1})$ and the predictive kernels $\ell_u(\theta'; J)$ induced by the same PBE.

The process $\{\mu_t\}_{t\geq 0}$ is **consistent with the PBE** if, under the ex-ante law $\mathbb{P}_{\mathrm{E}}$,

$$\mu_t(\theta) = \mathbb{P}_{\mathrm{E}}\Big(\theta_K = \theta \mid \mathcal{F}_t^{\mathrm{pub}}\Big) \qquad \forall\, \theta \in \Theta_K,\ \forall\, t \geq 1, \tag{S1.2}$$

interpreted $\mathbb{P}_{\mathrm{E}}$-almost surely upon fixing regular versions of the conditional probabilities.

□

Lemma S1.4 (Asymptotic stability of beliefs. Convergence theorem of Doob, 1953)**.** *Under the ex-ante law $\mathbb{P}_{\mathrm{E}}$ of Definition S1.3, the process $\{\mu_t(\theta)\}_{t\geq 0}$ is a **bounded martingale** that converges $\mathbb{P}_{\mathrm{E}}$-almost surely to a limit random variable $Z_\theta \in [0,1]$ $\mathbb{P}_{\mathrm{E}}$-a.s.*

Proof. The proof verifies the conditions of the convergence theorem of Doob (1953); Williams (1991); Shiryaev (2019).

1. Martingale property. Let $M = \mathbf{1}\{\theta_K = \theta\}$. By equilibrium consistency (Definition S1.3),

$$\mu_t(\theta) = \mathbb{E}^{\mathbb{P}_{\mathrm{E}}}[M \mid \mathcal{F}_t^{\mathrm{pub}}] \quad \mathbb{P}_{\mathrm{E}}\text{-a.s.}$$

As $\mathcal{F}_t^{\mathrm{pub}} \subseteq \mathcal{F}_{t+1}^{\mathrm{pub}}$, the **law of iterated expectations** yields

$$\mathbb{E}^{\mathbb{P}_{\mathrm{E}}}[\mu_{t+1}(\theta) \mid \mathcal{F}_t^{\mathrm{pub}}] = \mathbb{E}^{\mathbb{P}_{\mathrm{E}}}\Big[\mathbb{E}^{\mathbb{P}_{\mathrm{E}}}[M \mid \mathcal{F}_{t+1}^{\mathrm{pub}}] \mid \mathcal{F}_t^{\mathrm{pub}}\Big] = \mathbb{E}^{\mathbb{P}_{\mathrm{E}}}[M \mid \mathcal{F}_t^{\mathrm{pub}}] = \mu_t(\theta) \quad \mathbb{P}_{\mathrm{E}}\text{-a.s.}$$

The process is a martingale under $\mathbb{P}_{\mathrm{E}}$.

2. Pointwise and uniform L^1 bound. As $\mu_t(\theta)$ is a probability mass, $0 \leq \mu_t(\theta) \leq 1$ for all $t \geq 0$ (outside a $\mathbb{P}_{\mathrm{E}}$-null set). In particular, $\sup_{t\geq 0} \mathbb{E}^{\mathbb{P}_{\mathrm{E}}}[|\mu_t(\theta)|] \leq 1$, which provides the uniform L^1 bound required by Doob's theorem.

3. Almost sure convergence. Once the martingale property and the uniform L^1 bound are verified, **Doob's convergence theorem** guarantees the existence of Z_θ such that

$$\lim_{t\to\infty} \mu_t(\theta) = Z_\theta \quad \mathbb{P}_{\mathrm{E}}\text{-a.s.}$$

The pointwise bound $0 \le \mu_t(\theta) \le 1$ implies $Z_\theta \in [0,1]$ $\mathbb{P}_{\mathrm{E}}$-a.s. □

S2 Stochastic implementation mechanism (MDG)

This section establishes the fundamental definitions of the stochastic implementation layer (MDG), which transforms strategic intentions into materialized actions, and of the law governing the physical outcome that determines the observable results of the episode. The episode horizon is the endogenous stopping time

$$\tau := \inf\{t \ge 0 : m_t \in \mathcal{M}_{\mathrm{term}}\}, \tag{S2.1}$$

where m_t is the physical outcome, $\mathcal{M}_{\mathrm{cont}} = \{\mathrm{cont}\}$, $\mathcal{M}_{\mathrm{term}} = \{\mathrm{pay}, \mathrm{res}, \mathrm{rel}, \mathrm{kill}\}$, and $\mathcal{M}_{\mathrm{out}} = \mathcal{M}_{\mathrm{cont}} \cup \mathcal{M}_{\mathrm{term}}$ (fully formalized in Definition S2.4). Active periods satisfy $t < \tau$ while captivity continues.

Definition S2.1 (MDG stochastic implementation filter)**.** Let $i \in \{K, S, F\}$ and let $\mathcal{A}^i$ be the finite set of admissible actions of agent i. The **state vector** is $X_t := (\mu_t, h_t)$, with μ_t from Definition S1.3 and h_t the public history at the opening of t. Given the plan $a_t^{i*} \in \mathcal{A}^i$ and X_t, execution is determined by the **implementation law**

$$\tilde{a}_t^i \sim \mathbb{P}_{\mathrm{I},i}\Big(\tilde{a}_t^i \mid a_t^{i*},\, X_t\Big).$$

In the base logit specification (S2.3), $\mathbb{P}_{\mathrm{I},i}(\tilde{a}_t^i \mid a_t^{i*}, X_t)$ depends on X_t only through μ_t (via the temperature T_t); conditioning on h_t is reserved for extended specifications. The stochastic

structure is formalized on a measurable space $(\Omega, \mathcal{F})$. The measurable function $\mathcal{H}_t^i : \mathcal{A}^i \times \Omega \to \mathcal{A}^i$ materializes the sequential draw: $\tilde{a}_t^i(\omega) = \mathcal{H}_t^i(a_t^{i*}, \omega)$. The **aggregate profile of executed actions** is

$$\tilde{A}_t = (\tilde{a}_t^K,\ \tilde{a}_t^S,\ \tilde{a}_t^F). \tag{S2.2}$$

The following regularity assumptions are imposed.

(i) **Plan dominance and error support.** In the **base specification** with $\underline{c} > 0$, for all $t < \tau$ (with τ as in (S2.1)), $\mathbb{P}_{\mathrm{I},i}$ satisfies strict **plan dominance**, $\mathbb{P}_{\mathrm{I},i}(a_t^{i*} \mid a_t^{i*}, X_t) > \mathbb{P}_{\mathrm{I},i}(a \mid a_t^{i*}, X_t)$ for all $a \neq a_t^{i*}$, and **full support**, $\mathbb{P}_{\mathrm{I},i}(\tilde{a}_t^i = a \mid a_t^{i*}, X_t) > 0$ for all $a \in \mathcal{A}^i$.

(ii) **Bounded institutional maturation.** The informational component $[H(\mu_t)/H(\mu_0)]e^{-\eta_{\mathrm{cal}}t}$ declines as posterior uncertainty falls, while the structural floor $\underline{c}$ persists in (S2.4). When $\underline{c} > 0$, even if $H(\mu_t) \to 0$ one has $T_t \geq T_0\underline{c}$ and $\mathbb{P}_{\mathrm{I},i}(\tilde{a}_t^i = a_t^{i*} \mid a_t^{i*}, X_t) < 1$, so full support over $\mathcal{A}^i$ is preserved in the long run.

The implementation law adopts the **hybrid-temperature logit** structure:

$$\mathbb{P}_{\mathrm{I},i}\Big(\tilde{a}_t^i = a \mid a_t^{i*}, X_t\Big) = \frac{\exp\Big(\mathbf{1}\{a = a_t^{i*}\}/T_t\Big)}{\sum_{a' \in \mathcal{A}^i} \exp\Big(\mathbf{1}\{a' = a_t^{i*}\}/T_t\Big)}, \tag{S2.3}$$

with hybrid temperature

$$T_t = T_0 \max\Big\{\frac{H(\mu_t)}{H(\mu_0)} e^{-\eta_{\mathrm{cal}}t},\ \underline{c}\Big\}, \tag{S2.4}$$

where $T_0 > 0$ is the initial temperature, $\eta_{\mathrm{cal}} > 0$ is the calendar decay rate, Shannon entropy is $H(\mu_t) = -\sum_\theta \mu_t(\theta) \ln \mu_t(\theta)$ with $H(\mu_0) > 0$ by the full-support prior in Definition S1.3, and $\underline{c} \in [0, 1]$ is the floor of structural noise.

□

Remark S2.2 (Residual noise and the limit $\underline{c} = 0$)**.** The special case $\underline{c} = 0$ is interpreted as a limit without residual noise: it allows $\mathbb{P}_{\mathrm{E}}(\tilde{a}_t^i = a_t^{i*} \mid \mathcal{F}_t^{\mathrm{pub}}) \to 1$ when $H(\mu_t) \to 0$, but

sacrifices the persistent full support maintained under $\underline{c} > 0$. The base version with $\underline{c} > 0$ aligns with the absolute continuity or "grain of truth" condition of Kalai and Lehrer (1993), according to which rational learning must not assign zero probability to events that can occur.

Remark S2.3 (Axiomatic justification of the logit as the unique implementation law)**.** Equation (S2.3) is the *unique* distribution over $\mathcal{A}^i$ that simultaneously satisfies the three axioms:

(A1) **Full support:** $\mathbb{P}_{\mathrm{I},i}(a \mid a_t^{i*}, X_t) > 0$ for all $a \in \mathcal{A}^i$.

(A2) **Dominance:** $\mathbb{P}_{\mathrm{I},i}(a_t^{i*} \mid a_t^{i*}, X_t) > \mathbb{P}_{\mathrm{I},i}(a \mid a_t^{i*}, X_t)$ for all $a \neq a_t^{i*}$.

(A3) **Maximum entropy:** Among all distributions satisfying (A1) and (A2), $\mathbb{P}_{\mathrm{I},i}$ maximizes the Shannon entropy $H = -\sum_a p_a \ln p_a$ subject to the optimal action receiving fixed probability $1-\varepsilon_t \in (0,1)$, where $\varepsilon_t := 1 - \mathbb{P}_{\mathrm{I},i}(a_t^{i*} \mid a_t^{i*}, X_t)$ is the implementation-error rate induced by the logit.

In the parametrization (S2.3), ε_t is decreasing in $1/T_t$. By the characterization theorem of Jaynes (1957), the unique maximizer is the logit in (S2.3). The MDG process is thus axiomatically justified as the unique implementation process compatible with bounded rationality and maximum entropy.

Definition S2.4 (Law of the physical outcome)**.** The **physical outcome** m_t is an observable categorical variable on the support $\mathcal{M}_{\text{out}}$, where $\mathcal{M}_{\text{cont}} = \{\text{cont}\}$, $\mathcal{M}_{\text{term}} = \{\text{pay}, \text{res}, \text{rel}, \text{kill}\}$ and $\mathcal{M}_{\text{out}} = \mathcal{M}_{\text{cont}} \cup \mathcal{M}_{\text{term}}$. The label cont represents the continuation of captivity and $j \in \{1,2,3,4\}$ closure by competitive cause j. For $j \in \{1,2,3\}$, the effective intensity $\tilde{\lambda}_j(t \mid \theta_K)$ carries the type-specific parameter $\beta_{K,j}(\theta_K)$ in the Cox specification; cause $j = 4$ enters only through the baseline hazard (there is no $\beta_{K,4}$). Conditional on the captor's type

and the material state of the period

$$\mathcal{C}_t(\theta_K) = \Big(\theta_K, \theta_F, \theta_V, \theta_S, z, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, \alpha_t^*, \gamma_t^*, p_{\mathrm{det},t}\Big),$$

the distribution of m_t is defined from the effective intensities $\tilde{\lambda}_j(t \mid \theta_K)$. Denote by

$$Q_t(m \mid \mathcal{C}_t) := \mathbb{P}_{\mathrm{E}}(m_t = m \mid \mathcal{C}_t)$$

the transition kernel of the physical outcome, and by G_t the measurable rule that materializes a concrete realization of m_t from that kernel and a uniform draw. The **continuation probability** is

$$Q_t(\mathrm{cont} \mid \mathcal{C}_t) = \exp\left(-\sum_{j=1}^{4} \tilde{\lambda}_j(t \mid \theta_K)\, \Delta t\right).$$

The daily closure total and the relative weight of each cause are

$$q_t(\theta_K) = 1 - p_{\mathrm{Cont},t}, \quad \xi_j = \frac{\tilde{\lambda}_j}{\sum_\ell \tilde{\lambda}_\ell}, \quad \bar{h}_j(t \mid \theta_K) = q_t\, \xi_j.$$

The materialization of m_t is obtained by **inverse transform**: let $v_t \sim \mathrm{Unif}(0,1)$; then

$$m_t = r \iff \sum_{\ell<r} Q_t(\ell \mid \mathcal{C}_t) \le v_t < \sum_{\ell \le r} Q_t(\ell \mid \mathcal{C}_t),$$

and the likelihood of m_t in the Bayesian update is

$$\mathbb{P}_{\mathrm{E}}(m_t \mid \theta_K, \mathcal{C}_t) = p_{\mathrm{Cont}}^{\mathbf{1}\{m_t = \mathrm{cont}\}} \prod_{j=1}^{4} \bar{h}_j^{\mathbf{1}\{m_t = j\}}.$$

□

S3 Asymptotic type identification

This section defines the prolonged captivity event and the notion of asymptotic identification, states the structural identifiability lemma (bridge between the parameters $\beta_{K,j}$ and the singularity of laws), and concludes with the posterior consistency theorem.

Notation. Throughout Sections S3 and S7, θ_K^* denotes the realized captor type. When conditioning on the law of the true type, the notation $\theta^* := \theta_K^*$ and $\mu_C := \mathbb{P}_{C,\theta^*}(\cdot \mid \mathcal{T})$ is adopted.

Definition S3.1 (Prolonged captivity event)**.** Let τ be as in (S2.1). The **prolonged captivity event** is

$$\mathcal{T} := \{\tau = \infty\} = \Big\{\omega \in \Omega : m_t(\omega) \in \mathcal{M}_{\text{cont}} \; \forall\, t \geq 0\Big\}.$$

In $\mathcal{T}$ the episode admits no terminal outcome in any finite period and generates an infinite sequence of public signals $\{\zeta_t\}_{t\geq 0}$.

□

Definition S3.2 (Asymptotic identification in the event $\mathcal{T}$)**.** The model is **asymptotically identifiable** in $\mathcal{T}$ (Definition S3.1) if $\mathbb{P}_{C,\theta_K^*}(\mathcal{T}) > 0$ and, for every $\theta \in \Theta_K$ with $\theta \neq \theta_K^*$, at least one of the following conditions holds.

(a) **Exclusion by survival.** $\mathbb{P}_{C,\theta}(\mathcal{T}) = 0$.

(b) **Tail singularity of signals.** $\mathbb{P}_{C,\theta}(\mathcal{T}) > 0$ and the laws of $\{\zeta_t\}_{t\geq 0}$ restricted to $\mathcal{T}$ under $\mathbb{P}_{C,\theta}$ and under $\mathbb{P}_{C,\theta_K^*}$ are **mutually singular**.

□

Remark S3.3 (Role of Lemma S3.4 in the analysis)**.** Theorem S3.6 requires asymptotic identification (Definition S3.2), a condition formulated in terms of laws of entire processes.

Lemma S3.4 is the **operational bridge** between that abstract hypothesis and the parameters $\beta_{K,j}$ already introduced in the mechanism: it allows one to verify in the practice of the model when two distinct types generate asymptotically distinguishable trajectories. Without this result, the learning section would rest on a high-level condition difficult to connect with the proportional parametrization of the λ_j.

Lemma S3.4 (Structural identifiability and asymptotic learning)**.** *Suppose* $|\Theta_K| < \infty$. *Let* $Y_t := (m_t, d_t)$ *be the daily public record. Under a PBE,* $p_\theta(\cdot \mid \mathcal{F}_t^{\text{pub}})$ *denotes the mass function of* Y_t *conditioned on public information at the opening of* t *under type* θ.

Part (i) — local sufficiency (Cox, 1972; Bradburn et al., 2003). *Let* $\theta, \theta' \in \Theta_K$ *be distinct. If there exists* $j \in \{1,2,3\}$ *such that* $\beta_{K,j}(\theta) \neq \beta_{K,j}(\theta')$ *(causes* $j \in \{1,2,3\}$ *only; see Definition S2.4) and no exact compensations arise between the masses of* Y_t *(*knife-edges*), then there exists* $\varepsilon = \varepsilon(\theta, \theta', t) > 0$ *such that*

$$\left\| p_\theta(\cdot \mid \mathcal{F}_t^{\text{pub}}) - p_{\theta'}(\cdot \mid \mathcal{F}_t^{\text{pub}}) \right\|_{\text{TV}} \geq \varepsilon. \tag{S3.1}$$

Part (ii) — asymptotic sufficiency (Kakutani, 1948; Shiryaev, 2019). *Fixing a policy trajectory* $(\chi_u)_{u \geq 0}$, *suppose that conditional on* $\mathcal{T}$ *the* $\{Y_t\}_{t \geq 0}$ *are independent. For each pair* $\theta \neq \theta'$, *let* $\mathcal{S}_{\theta,\theta'} \subseteq \mathbb{N}$ *be the set of periods where* (S3.1) *holds. If* $\mathcal{S}_{\theta,\theta_K^*}$ *is infinite* $\mathbb{P}_{\text{C},\theta_K^*}$*-a.s. and there exists* $\varepsilon_\theta > 0$ *such that* $\|p_\theta^{(t)} - p_{\theta_K^*}^{(t)}\|_{\text{TV}} \geq \varepsilon_\theta$ *for all* $t \in \mathcal{S}_{\theta,\theta_K^*}$, *then the laws of* $(Y_t)_{t \geq 0}$ *conditioned on* $\mathcal{T}$ *under* $\mathbb{P}_{\text{C},\theta}$ *and under* $\mathbb{P}_{\text{C},\theta_K^*}$ *are* ***mutually singular***.

Remark S3.5 (Informational Slicing and Conditional Independence)**.** A crucial conceptual distinction must be maintained between the adaptive equilibrium path of the active dual control policy and the measure-theoretic construction required by Lemma S3.4 (Kakutani, 1948; Blackwell and Dubins, 1962). In the live perfect Bayesian equilibrium, the planner updates beliefs sequentially, causing the realized instruments (α_t^*, γ_t^*) to be non-trivially coupled with the historical filtration $\{\mathcal{F}_t^{\text{pub}}\}$ (Aghion et al., 1991; Easley and Kiefer, 1988).

However, the probability measures $\mathbb{P}_{C,\theta}$ invoked in Definition S3.2 and Lemma S3.4 operate on a frozen deterministic policy path $(\chi_u)_{u\geq 0}$ evaluated ex-ante at $t = 0$ (Blackwell and Dubins, 1962). Within this sliced counterfactual regime, the strategic intentions of the private agents are uniquely pinned down by sequential rationality, rendering the daily innovations of the physical-institutional block $Y_t = (m_t, d_t)$ conditionally independent across periods (Kakutani, 1948). This conditional factorization satisfies the orthogonal product-measure structure $\left(\bigotimes_{t\geq 0} p_\theta^{(t)}\right)$ required to validly operationalize the singularity criterion of Kakutani (1948) without violating the temporal feedback loop of the dynamic game.

Proof. The proof identifies the causal mechanism within the day, translates it to a statistical bound in total variation, and couples the repeated experiment to the criterion of Kakutani (1948).

1. Same control menu, distinct criminal technology. Fix t and $\mathcal{F}_t^{\text{pub}}$. In the counterfactual that changes only the type (θ vs. θ'), keeping equal the executed triple $\tilde{A}_t$, the instruments (α_t^*, γ_t^*), and the maturation filter $M(t)$, the proportional parametrization implies

$$\frac{\lambda_j(t \mid \theta)}{\lambda_j(t \mid \theta')} = \exp\Big(\beta_{K,j}(\theta) - \beta_{K,j}(\theta')\Big) \neq 1.$$

2. From daily intensity to distinct predictions over Y_t. The proportional difference in $\lambda_j(t)$ is projected onto the aggregate exponential of $q(t)$, the causal mix $\xi_j(t)$, or the marginal hazards $\bar{h}_j(t) = q(t)\xi_j(t)$. Except for *knife-edges*, the marginal of m_t cannot coincide under θ and θ', and the two mass functions differ in total variation, verifying (S3.1).

3. Repetition with uniform margin and Kakutani criterion. Conditional on the fixed policy path and on $\mathcal{T}$—under the informational slicing regime clarified in Remark S3.5—the law of the sequence factors exactly as a product of daily independent experiments $\bigotimes_{t\geq 0} p_\theta^{(t)}$ versus $\bigotimes_{t\geq 0} p_{\theta_K^*}^{(t)}$ (Kakutani, 1948). Part (ii) then follows directly from Lemma S7.1 (Section S7): recurrent uniform total-variation separation on an infinite set forces the Bhattacharyya product affinity to zero and, by the criterion of Kakutani (1948), mutual singu-

larity of the infinite product measures (Shiryaev, 2019). □

Theorem S3.6 (Identification and consistency of true types)**.** *Consider a prior $\mu_0 \in \Delta(\Theta_K)$ with full support and a PBE-consistent Bayesian updating (Definition S1.3). Let $\theta^* := \theta^*_K$ be the realized type and $\mu_C := \mathbb{P}_{C,\theta^*}(\cdot \mid \mathcal{T})$ the law of the true type restricted to the event $\mathcal{T}$ (Definition S3.1).*

If the mechanism is ***asymptotically identifiable*** *in $\mathcal{T}$ (Definition S3.2), the following properties are satisfied μ_C-almost surely:*

1. ***Exclusion of false hypotheses:*** $\mu_t(\theta) \xrightarrow{t\to\infty} 0$ *for all* $\theta \neq \theta^*$.
2. ***Concentration on the true type:*** $\mu_t(\theta^*) \xrightarrow{t\to\infty} 1$.

Consequently, in $\mathcal{T}$ the sequence $\{\mu_t\}$ converges weakly to the Dirac measure δ_{θ^*}.

Proof. The proof separates the general probabilistic argument from the structural identification assumption.

S3.0.0.1 1. Stability of beliefs. By on-path Bayesian consistency of the PBE (condition (b) of Definition S6.1) and Definition S1.3, the belief process is represented as $\mu_t(\theta) = \mathbb{E}^{\mathbb{P}_E}[\mathbf{1}\{\theta_K = \theta\} \mid \mathcal{F}^{\text{pub}}_t]$ $\mathbb{P}_E$-a.s. Lemma S1.4 guarantees that $\{\mu_t(\theta)\}$ is a bounded martingale under $\mathbb{P}_E$ with limit $Z_\theta \in [0,1]$ $\mathbb{P}_E$-a.s. The transfer to μ_C-a.s. is justified by disintegration: the mixture $\mathbb{P}_E = \sum_\vartheta \mu_0(\vartheta)\mathbb{P}_{C,\vartheta}$ implies that every $\mathbb{P}_E$-null set is $\mathbb{P}_{C,\theta^*}$-null (since $\mu_0(\theta^*) > 0$), and domination is preserved upon restricting to the prolonged captivity event $\mathcal{T}$ (Definition S3.1).

S3.0.0.2 2. Identification and exclusion of false types. Fix $\theta \neq \theta^*$.

Case $\mathbb{P}_{C,\theta}(\mathcal{T}) = 0$: the trajectory under μ_C belongs to an event that the false type cannot generate; its posterior likelihood is null in the conditioned regime.

Case $\mathbb{P}_{C,\theta}(\mathcal{T}) > 0$: the singularity of Definition S3.2(b) implies that there exists a tail event A_θ with $\mathbb{P}_{C,\theta^*}(A_\theta \mid \mathcal{T}) = 1$ and $\mathbb{P}_{C,\theta}(A_\theta \mid \mathcal{T}) = 0$. Let R^θ_t be the likelihood ratio of the

false type against the true type on finite horizons. Tail singularity forces $R_t^\theta \to 0$ μ_C-a.s. (Blackwell and Dubins, 1962; Schervish, 1995). By the Bayesian updating rule defined in Definition S1.3 with a full-support prior, the posterior belief satisfies

$$\mu_t(\theta) = \frac{\mu_0(\theta)\, R_t^\theta}{\mu_0(\theta^*) + \sum_{\theta' \neq \theta^*} \mu_0(\theta') R_t^{\theta'}} \leq \frac{\mu_0(\theta)}{\mu_0(\theta^*)}\, R_t^\theta,$$

and since $\mu_0(\theta^*) > 0$, it follows that $\mu_t(\theta) \to 0$ μ_C-a.s. for all $\theta \neq \theta^*$.

S3.0.0.3 3. Closure by normalization. For all t, $\sum_{\theta \in \Theta_K} \mu_t(\theta) = 1$. As Θ_K is finite and all false coordinates converge to zero μ_C-a.s.,

$$1 = \lim_{t \to \infty} \sum_\theta \mu_t(\theta) = Z_{\theta^*} + \sum_{\theta \neq \theta^*} 0,$$

hence $Z_{\theta^*} = 1$ μ_C-a.s., that is, $\mu_t(\theta^*) \to 1$ μ_C-a.s. On a finite simplex, coordinate convergence is equivalent to weak convergence $\mu_t \to \delta_{\theta^*}$. □

Remark S3.7 (Economic reading and link with $\beta_{K,j}$)**.** Definition S3.2 requires that the design of the game (policy + incentives + risk technology) produce long trajectories incompatible with false types. Distinct shifts $\beta_{K,j}(\theta) \neq \beta_{K,j}(\theta')$ favor the singularity of the laws of (m_t, d_t). If the equilibrium pools types, Lemma S1.4 still holds but Theorem S3.6 does not: the limit may split mass among observationally equivalent types.

Remark S3.8 (Finite horizon)**.** If $\tau < \infty$ almost surely, $\mathcal{T}$ has measure zero and the statement of Theorem S3.6 is vacuous. The relevant conclusion is then only the martingale convergence of Lemma S1.4: beliefs stabilize at the finite limit of observations.

S4 Competitive risks and operational pressure

The following proposition uses directly the proportional risk structure, without requiring the equilibrium machinery of Section S6. It depends only on the intensities $\tilde{\lambda}_j$ introduced in

Definition S2.4.

Proposition S4.1 (Expected duration of captivity and operational pressure)**.** *Fix a type θ_K and a period t. The following standing assumptions govern the comparative static in γ_t^*.*

Standing assumptions. (S0) *Baseline hazards satisfy $\lambda_{j0}(s) > 0$ for all strategic causes $j \in \{1,2,3\}$ and all s in the support, so $\mathbb{E}[\tau \mid \theta_K] < \infty$.* (S1) *The object $\partial \mathbb{E}[\tau \mid \theta_K]/\partial \gamma_t^*$ is a partial derivative: only γ_t^* is perturbed at period t while $\{\gamma_u^*\}_{u \neq t}$ is held fixed. Writing $S(u \mid \theta_K) := \prod_{r=1}^{u} p_{\mathrm{Cont},r|\theta_K}$ with $S(0 \mid \theta_K) = 1$, expected duration admits the survival representation*

$$\mathbb{E}[\tau \mid \theta_K] = \sum_{u=0}^{\infty} S(u \mid \theta_K), \tag{S4.1}$$

so only factors with $u \geq t$ depend on γ_t^*. (S2) *For each strategic cause $j \in \{1,2,3\}$,*

$$\frac{\partial \tilde{\lambda}_j(t \mid \theta_K)}{\partial \gamma_t^*} = \eta_{\gamma,j}\, \tilde{\lambda}_j(t \mid \theta_K), \qquad \eta_{\gamma,j} := \frac{\partial \ln \tilde{\lambda}_j(t \mid \theta_K)}{\partial \gamma_t^*};$$

the exogenous cause $j = 4$ is invariant to γ_t^*.

Net sensitivity. In the main-paper hazard index, payment ($j = 1$) is decreasing in γ and death/state rescue ($j \in \{2,3\}$) are increasing, so $\eta_{\gamma,1} < 0$ and $\eta_{\gamma,j} > 0$ for $j \in \{2,3\}$. Let $\zeta_{\gamma,j} := |\eta_{\gamma,j}|$ and define

$$\kappa_h(\theta_K, t) := \zeta_{\gamma,2}\, \tilde{\lambda}_2(t \mid \theta_K) + \zeta_{\gamma,3}\, \tilde{\lambda}_3(t \mid \theta_K) - \zeta_{\gamma,1}\, \tilde{\lambda}_1(t \mid \theta_K) = \sum_{j=1}^{3} \eta_{\gamma,j}\, \tilde{\lambda}_j(t \mid \theta_K). \tag{S4.2}$$

Result. Under (S0)–(S2)*,*

$$\frac{\partial\, \mathbb{E}[\tau \mid \theta_K]}{\partial \gamma_t^*} = -\kappa_h(\theta_K, t)\, \Delta t \sum_{u=t}^{\infty} S(u \mid \theta_K). \tag{S4.3}$$

In particular, whenever $\kappa_h(\theta_K, t) \neq 0$,

$$\operatorname{sgn}\left(\frac{\partial\, \mathbb{E}[\tau \mid \theta_K]}{\partial \gamma_t^*} \right) = -\operatorname{sgn}(\kappa_h(\theta_K, t))\,. \tag{S4.4}$$

If $\kappa_h(\theta_K, t) > 0$, the one-shot perturbation in (S1) *lowers expected duration; if $\kappa_h(\theta_K, t) < 0$, it raises it. Cross-type variation in the sign enters through $\beta_{K,j}(\theta_K)$ and the induced semi-elasticities $\{\eta_{\gamma,j}\}$.*

Proof. The argument has three steps: differentiate the daily continuation probability, propagate the perturbation through the survival product, and differentiate the duration identity (S4.1).

S4.0.0.1 Step 0. Survival representation. For a discrete-time competing-risks process with daily continuation probability $p_{\text{Cont},s|\theta_K}$, the survival function $S(t' \mid \theta_K) = \prod_{s=1}^{t'} p_{\text{Cont},s|\theta_K}$ satisfies $S(t' \mid \theta_K) = \mathbb{P}(\tau > t' \mid \theta_K)$. Hence

$$\mathbb{E}[\tau \mid \theta_K] = \sum_{t'=0}^{\infty} \mathbb{P}(\tau > t' \mid \theta_K) = \sum_{t'=0}^{\infty} S(t' \mid \theta_K),$$

which is the discrete-time survival identity (Wald's representation; see, e.g., Kalbfleisch and Prentice (2002)). Finiteness in (S0) makes the sum well defined.

S4.0.0.2 Step 1. Daily continuation and κ_h. By Definition S2.4,

$$p_{\text{Cont},s|\theta_K} = \exp\Big(-\sum_{j=1}^{4} \tilde{\lambda}_j(s \mid \theta_K)\,\Delta t\Big) = \exp\Big(-[\Lambda_s(\theta_K) + \tilde{\lambda}_4(s)]\,\Delta t\Big),$$

where $\Lambda_s(\theta_K) := \sum_{j=1}^{3} \tilde{\lambda}_j(s \mid \theta_K)$ is the strategic hazard total. Only Λ_t depends on γ_t^*; the exogenous term $\tilde{\lambda}_4(t)$ is γ-invariant. By (S2),

$$\frac{\partial \Lambda_t(\theta_K)}{\partial \gamma_t^*} = \sum_{j=1}^{3} \eta_{\gamma,j}\, \tilde{\lambda}_j(t \mid \theta_K) = \kappa_h(\theta_K, t),$$

using $\eta_{\gamma,1} = -\zeta_{\gamma,1}$ and $\eta_{\gamma,j} = +\zeta_{\gamma,j}$ for $j \in \{2, 3\}$ in (S4.2). Since $p_{\text{Cont},t|\theta_K} = \exp(-\Lambda_t \Delta t) \cdot \exp(-\tilde{\lambda}_4(t)\Delta t)$ and the second factor is constant in γ_t^*,

$$\frac{\partial\, p_{\text{Cont},t|\theta_K}}{\partial \gamma_t^*} = \exp(-\tilde{\lambda}_4(t)\Delta t) \cdot \frac{\partial}{\partial \gamma_t^*} \exp(-\Lambda_t \Delta t) = -\kappa_h(\theta_K, t)\; p_{\text{Cont},t|\theta_K}\; \Delta t. \tag{S4.5}$$

S4.0.0.3 Step 2. Propagation to accumulated survival. Fix $t' \geq t$. Because only the factor $s = t$ in $S(t' \mid \theta_K) = \prod_{s=1}^{t'} p_{\text{Cont},s|\theta_K}$ depends on γ_t^*,

$$\frac{\partial S(t' \mid \theta_K)}{\partial \gamma_t^*} = \Big(\prod_{\substack{s=1 \\ s \neq t}}^{t'} p_{\text{Cont},s|\theta_K}\Big) \frac{\partial\, p_{\text{Cont},t|\theta_K}}{\partial \gamma_t^*} = -\kappa_h(\theta_K, t)\; S(t' \mid \theta_K)\; \Delta t.$$

For $t' < t$, the derivative is zero.

S4.0.0.4 Step 3. Expected duration and sign. Differentiating (S4.1) term by term under (S1) and applying Step 2 yields (S4.3). The interchange of sum and derivative is justified by dominated convergence: for $t' \geq t$, $|\partial S(t' \mid \theta_K)/\partial \gamma_t^*| \leq |\kappa_h(\theta_K, t)|\, S(t' \mid \theta_K)\, \Delta t$, and

$$\sum_{t'=t}^{\infty} S(t' \mid \theta_K) \leq \sum_{t'=0}^{\infty} S(t' \mid \theta_K) = \mathbb{E}[\tau \mid \theta_K] < \infty$$

by (S0). Because $S(t' \mid \theta_K) \geq 0$ and the tail $\sum_{t'=t}^{\infty} S(t' \mid \theta_K) > 0$ whenever continuation has positive mass on $t' \geq t$, the sign of (S4.3) is the opposite of the sign of $\kappa_h(\theta_K, t)$. □

Remark S4.2 (Local comparative static and testable content)**.** Proposition S4.1 is a *local* comparative static: it evaluates a one-period perturbation of γ_t^* holding the remaining policy path fixed. It does not claim global monotonicity of $\mathbb{E}[\tau \mid \theta_K]$ as a function of the entire sequence $\{\gamma_s^*\}_{s \geq 0}$. The empirical content is the sign prediction: types with larger κ_h respond differently to operational pressure because the Cox coefficients $\beta_{K,j}(\theta_K)$ shift the semi-elasticities $\{\eta_{\gamma,j}\}$ and the levels $\{\tilde{\lambda}_j\}$.

S5 Policy implications

This proposition requires the belief machinery of Section S1 and the equilibrium context of Section S6: optimal pressure is calibrated in a Bayesian manner.

Proposition S5.1 (Optimal operational pressure and Bayesian type identification)**.** *Suppose* $\Theta_K = \{\theta_{low}, \theta_{high}\}$ *and fix a discrete branch* $b \in \{R, N\}$ *in a neighborhood of the posterior*

belief under consideration. Let $x_t = (\alpha_t, \gamma_t)$ *and define the complete branch objective*

$$J_t^b(x_t, \mu_t) := \sum_{\theta \in \Theta_K} \mu_t(\theta) L_t(b, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi_{t,b}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t). \tag{S5.1}$$

If (α_t^*, γ_t^*) *is an interior optimum of* $J_t^b(\cdot, \mu_t)$*, the constraints* IC^K*,* IR^K*, and* IR^F *do not locally change regime, the Hessian* $H_t^b := \nabla_{xx}^2 J_t^b(x_t^*, \mu_t)$ *is positive definite, and the net marginal effect of increasing* $\mu_t(\theta_{high})$ *on the first-order condition for* γ_t *favors greater pressure, that is,*

$$e_\gamma' \Big(H_t^b\Big)^{-1} \nabla_{x\mu}^2 J_t^b(x_t^*, \mu_t) < 0, \qquad e_\gamma = (0,1)', \tag{S5.2}$$

then the optimal operational pressure of that branch is locally increasing in the posterior belief on the high-risk type:

$$\frac{d\gamma_t^*}{d\mu_t(\theta_{high})} > 0.$$

The structural separation $\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})$ *for* $j \in \{2,3\}$ *contributes to this condition through the ratio*

$$\frac{\tilde{\lambda}_j(t \mid \theta_{high})}{\tilde{\lambda}_j(t \mid \theta_{low})} = \exp\Big(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})\Big), \quad j \in \{2,3\}.$$

Monotonicity, however, depends on the joint net effect of material losses, deviation penalties, and informational gains.

Proof. Write $\mu := \mu_t(\theta_{high})$ and $\mu_t(\theta_{low}) = 1 - \mu$. Within the fixed discrete branch b, the interior optimum $x_t^*(\mu) = (\alpha_t^*(\mu), \gamma_t^*(\mu))$ satisfies

$$\nabla_x J_t^b(x_t^*(\mu), \mu) = 0. \tag{S5.3}$$

The hypothesis $H_t^b = \nabla_{xx}^2 J_t^b(x_t^*, \mu) \succ 0$ implies that the local optimum is isolated and that the Jacobian of the first-order condition with respect to x_t is invertible. By the implicit

function theorem, $x_t^*(\mu)$ is locally differentiable and

$$\frac{dx_t^*}{d\mu} = -\Big(H_t^b\Big)^{-1}\nabla_{x\mu}^2 J_t^b(x_t^*, \mu). \tag{S5.4}$$

Taking the second coordinate yields

$$\frac{d\gamma_t^*}{d\mu_t(\theta_{high})} = -e_\gamma'\Big(H_t^b\Big)^{-1}\nabla_{x\mu}^2 J_t^b(x_t^*, \mu_t). \tag{S5.5}$$

By the net monotonicity condition (S5.2), the right-hand side of (S5.5) is strictly positive.

To interpret this condition, note that the coordinate of $\nabla_{x\mu}^2 J_t^b$ associated with γ_t contains

$$\frac{\partial L_t}{\partial \gamma_t}(b, \alpha_t^*, \gamma_t^*, \theta_{high}, \iota_t) - \frac{\partial L_t}{\partial \gamma_t}(b, \alpha_t^*, \gamma_t^*, \theta_{low}, \iota_t) + \frac{\partial}{\partial \mu}\left[\frac{\partial \Pi_{t,b}^S}{\partial \gamma_t} - \psi_H \frac{\partial \Delta H_t}{\partial \gamma_t}\right]_{x_t = x_t^*}. \tag{S5.6}$$

The first differential captures the material separation between types. Under proportional risks,

$$\frac{\tilde{\lambda}_j(t \mid \theta_{high})}{\tilde{\lambda}_j(t \mid \theta_{low})} = \exp\Big(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})\Big),$$

so that greater structural distance in $j \in \{2, 3\}$ increases the difference in marginal losses associated with operational pressure. However, the deviation premium and the informational subsidy also enter the first-order condition; therefore the result is formulated as monotonicity of the net marginal effect of the complete objective, not as a property of material loss alone. Under (S5.2), increasing the posterior mass on the high-risk type shifts the interior optimum toward a higher γ_t^*. □

S6 Equilibrium and implementability

This section introduces the dynamic mechanism with its PBE, the notion of implementability with the constraints IC^K, IR^K, IR^F, and proves the existence of the implementable equilibrium.

Definition S6.1 (Assessment and Perfect Bayesian Equilibrium)**.** Consider a discrete-time Bayesian game with horizon $\mathbb{T} = \{0, 1, \ldots, \tau\}$, players $\mathcal{J} = \{S, K, F\}$, and captor type $\theta_K \in \Theta_K$ according to common prior μ_0 with $\mu_0(\theta) > 0$ for all θ. An **assessment** in this dynamic game is described by:

- *Strategy profile:* $s = (s^S, s^K, s^F)$, where $s^S = \chi = (\chi_t)_{t\in\mathbb{T}}$ with $\chi_t : H_t \times \mathcal{R} \to \mathcal{A}^{S,\text{disc}} \times \mathcal{A}^{\alpha} \times \mathcal{A}^{\gamma}$.
- *Belief system:* $\mu = \{\mu_t\}_{t\geq 0}$, $\mu_t \in \Delta(\Theta_K)$.
- *Ex-ante law:* $\mathbb{P}^s_{\mathrm{E}}$ induced by profile s, prior μ_0, and the physical dynamics of captivity.

Perfect Bayesian Equilibrium. The pair (s, μ) is a **PBE** if the following hold:

(a) **Sequential rationality.** At every node reached with positive probability, each agent i solves its optimization program given μ_t and s^{-i}. For the state:

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg\min_{(a_t^S, \alpha_t, \gamma_t)} \Big\{ \sum_{\theta} \mu_t(\theta) L_t(a_t^S, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi^S_{t,a^S}(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \Big\}.$$

For $i \in \{K, F\}$: $a_t^{i*} \in \arg\max_{a_t^i} \mathcal{U}_t^i(a_t^i \mid \mathcal{I}_t^i, \chi, s^{-i})$.

(b) **On-path Bayesian consistency.**

$$\mu_t(\theta) = \mathbb{P}^s_{\mathrm{E}}(\theta_K = \theta \mid \mathcal{F}_t^{\text{pub}}), \qquad \forall\, \theta,\ t \geq 1.$$

□

Remark S6.2. Definition S6.1 defines the PBE as an *object*: the pair (s, μ) satisfying (a) and (b). It does not assert existence nor that the profile satisfies incentive and participation constraints; those properties are the content of Theorem S6.6.

Definition S6.3 (Nash-Bayes-IC^K,IR^K,IR^F implementability)**.** Let (s^*, μ^*) be a PBE (Definition S6.1). The mechanism is **implementable Nash-Bayes-IC^K,IR^K,IR^F** if the profile additionally satisfies the following conditions:

Bayesian incentive compatibility of the captor (IC^K). For all $t \in \mathbb{T}$ and all rival mimicry $\theta_j \in \Theta_K$,

$$\mathbb{E}_{\theta\sim\mu_t^*}\Big[V_t^K\Big(a^*(\theta) \mid \theta, \alpha_t^*, \gamma_t^*\Big) - V_t^K\Big(a^*(\theta_j) \mid \theta, \alpha_t^*, \gamma_t^*\Big)\Big] \geq 0.$$

Participation constraint of the captor (IR^K).

$$\mathbb{E}_{\theta\sim\mu_t^*}\Big[U_{\text{rel}}^K(\theta) - \max\Big\{V_{\text{cont},t}^K(\theta, \alpha_t^*, \gamma_t^*), U_{\text{kill}}^K(\theta, \theta_S)\Big\}\Big] \geq 0.$$

Participation constraint of the family (IR^F).

$$\mathcal{U}_t^F(a_{\text{coop}}) \geq \mathcal{U}_t^F(a_{\text{col}}).$$

The **feasible set of policies** is

$$\Gamma_t(\mu_t) := \Big\{(a_t^S, \alpha_t, \gamma_t) \in \mathcal{A}^{S,\text{disc}} \times [0,1]^2 : IC^K,\ IR^K,\ IR^F \text{ are satisfied}\Big\}. \tag{S6.1}$$

Its non-emptiness — **robust feasibility** — is the primitive hypothesis of Theorem S6.6.

□

Remark S6.4. Definition S6.3 states implementability as a *property* of the pair (s^*, μ^*): given the existence of the PBE, one asks whether it additionally satisfies IC^K, IR^K, and IR^F. The cycle criterion of Rochet (1987) provides a useful algebraic reformulation for auditing IC^K when Θ_K is finite.

Remark S6.5 (Full support of the MDG and Bayesian operability)**.** The full support of $\mathbb{P}_{\text{I},i}$ (hypothesis (iii) of Theorem S6.6) guarantees strictly positive probability at all reachable

nodes, making Bayesian updating operative on the path and reducing indeterminacy off the path.

Theorem S6.6 (Conditional implementability of the mechanism in PBE)**.** *Consider the PBE of Definition S6.1 under the following conditions:*

(i) $|\Theta_K| < \infty$ *with strictly positive prior.*

(ii) *Finite action spaces* $\mathcal{A}^i$ *for all* $i \in \{K, S, F\}$.

(iii) *The law* $\mathbb{P}_{\mathrm{I},i}$ *of the MDG process assigns strictly positive probability to every action (**full support**).*

(iv) $\mathcal{A}^\alpha = \mathcal{A}^\gamma = [0, 1]$, *compact and convex;* L_t *and* $\mathcal{U}_t^i$ *are measurable, bounded, and continuous in the continuous decision variables. The physical outcome rule* G_t *is measurable, and the transition kernel* Q_t *induced by* G_t *and by the competitive risk probabilities is continuous in those variables.*

(v) ***Robust feasibility.*** *For each* t *and* $\mu_t \in \Delta(\Theta_K)$, $\Gamma_t(\mu_t)$ *is non-empty and compact; the sections* $\Gamma_t^{a^S}(\mu_t)$ *are convex; the correspondence* $\mu_t \mapsto \Gamma_t(\mu_t)$ *is continuous with compact values.*

(vi) ***Parametric separability.***

$$V_t^K(a^*(\theta') \mid \theta_K, \alpha_t, \gamma_t) = \sum_j w_j(a^*(\theta'), \alpha_t, \gamma_t) \exp(\beta_{K,j}(\theta_K) \cdot z).$$

(vii) ***Quasiconvexity of the restricted objective.*** *For each* t, μ_t, *and* a_t^S, *the functional* $(\alpha_t, \gamma_t) \mapsto \sum_\theta \mu_t(\theta) L_t + \Pi_{t,a^S}^S - \psi_H \Delta H_t$ *is quasiconvex in* (α_t, γ_t) *over* $\Gamma_t^{a^S}(\mu_t)$.

Then there exists (s^*, μ^*) *such that:*

(a) *Constitutes a **PBE** (Definition S6.1).*

(b) ***Implements the social choice rule*** f_t*:* $\chi_t^*(h_t) = (a_t^{S*}, \alpha_t^*, \gamma_t^*)$ *minimizes the expected social loss subject to* $\Gamma_t(\mu_t^*)$.

(c) ***Satisfies*** IC^K***,*** IR^K***, and*** IR^F *in profile* (s^*, μ^*).

Proof. The proof proceeds in five steps.

S6.0.0.1 1. Existence of equilibrium. By (ii), the behavioral strategy space Σ^i of each agent defined in the assessment (Definition S6.1) is a finite product of simplices, compact and convex. By (v), the feasible set of policies $\Gamma_t(\mu_t)$ defined in Definition S6.3 is compact, non-empty, with convex sections and continuous correspondence. The state's objective in the sequential rationality condition (a) of Definition S6.1 is continuous in (α_t, γ_t) by (iv); by **Berge's maximum theorem** (Aliprantis and Border, 2006), the correspondence of optima $\Psi_t(\mu_t)$ is upper hemicontinuous and compact. By (vii), the functional is quasiconvex; the argmin of a quasiconvex function over a compact convex set is convex, so $\Psi_t(\mu_t)$ has convex values. The utilities of K and F from Definition S6.1 are multilinear in behavioral strategies, so the best response correspondence

$$\mathrm{BR}(s) := \mathrm{BR}_\Gamma^S(s^{-S}) \times \mathrm{BR}^K(s^{-K}) \times \mathrm{BR}^F(s^{-F})$$

has non-empty convex values. By **Kakutani's fixed-point theorem** (Kakutani, 1941), there exists s^* with $s^* \in \mathrm{BR}(s^*)$.

S6.0.0.2 2. On-path Bayesian consistency. By (iii), $\mathbb{P}_{\mathrm{I},i}$ assigns positive probability to all actions compatible with the physical dynamics. Bayes' formula

$$\mu_t^*(\theta) = \mathbb{P}_{\mathrm{E}}^{s^*}\Big(\theta_K = \theta \mid \sigma(z, \zeta_0, \ldots, \zeta_{t-1})\Big)$$

is well defined at reached nodes and satisfies condition (b) of Definition S6.1.

S6.0.0.3 3. Implementation of the social choice rule. At the fixed point, the state's strategy solves at each node on the path, satisfying the sequential rationality condition (a) of Definition S6.1:

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{(a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t^*)} \Big\{ \sum_\theta \mu_t^*(\theta) L_t + \Pi^S_{t,a^S}(\alpha_t, \gamma_t; \mu_t^*) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \Big\}.$$

The objective is continuous by (iv) and the feasible set $\Gamma_t(\mu_t^*)$ (Definition S6.3) is compact and non-empty by (v); by Weierstrass a solution exists.

S6.0.0.4 4. Incentive compatibility (IC^K): cycle reformulation (Rochet, 1987). IC^K is imposed through the feasible set $\Gamma_t(\mu_t^*)$, not verified ex post: equilibrium instruments belong to $\Gamma_t(\mu_t^*)$, which by (v) and Definition S6.3 incorporates IC^K as a constitutive condition. For algebraic auditing with $\Delta^t_{ij} := V_t^K(a^*(\theta_j) \mid \theta_i) - V_t^K(a^*(\theta_i) \mid \theta_i)$, the Bayesian version is equivalent to $\mathbb{E}_{\theta_i \sim \mu_t^*}[\Delta^t_{ij}] \leq 0$ for all j. For every cycle $(\theta_{k_0}, \ldots, \theta_{k_m}, \theta_{k_0})$:

$$\sum_{\ell=0}^{m} \mu_t^*(\theta_{k_\ell})\, \Delta^t_{k_\ell, k_{\ell+1}} \leq 0 \qquad \text{(indices mod } m+1\text{)}.$$

S6.0.0.5 5. Participation constraints (IR^K, IR^F). By construction in Step 3, the equilibrium instruments satisfy $(\alpha_t^*, \gamma_t^*) \in \Gamma_t(\mu_t^*)$, which by Definition S6.3 simultaneously guarantees that the participation constraints IR^K and IR^F are satisfied in the equilibrium profile.

Having completed the five steps, (s^*, μ^*) is a PBE (Definition S6.1) that implements the social choice rule f_t and is implementable in the sense of Definition S6.3 (satisfying IC^K, IR^K, and IR^F simultaneously). □

Remark S6.7 (Separation of results: implementability and asymptotic identification)**.** Theorem S6.6 and Theorem S3.6 guarantee complementary properties. The first guarantees existence of a PBE with implementable policy subject to Γ_t: IC^K enters through the definition of the feasible set. The second guarantees that under that same equilibrium, if heterogeneity

in $\beta_{K,j}$ generates uniform separation in total variation, beliefs converge to the true type almost surely in $\mathcal{T}$. The full support of the MDG plays a role in both: it reduces the *off-path* region and avoids degeneracies of the likelihood.

S7 Abstract structural identification (Kakutani criterion)

This section isolates the measure-theoretic argument: under product structure and recurrent separation in total variation, the criterion of Kakutani (1948) forces mutual singularity of trajectory laws. Theorem S7.2 states the same Bayesian conclusion as Section S3, replacing Lemma S3.4 with the abstract hypotheses of the following lemma.

Lemma S7.1 (Product of experiments and singularity (Kakutani criterion))**.** *Fix $\theta \neq \theta'$. Suppose that conditional on equilibrium policy and on event $\mathcal{T}$, the sequence $\{Y_t\}_{t\geq 0}$ is independent and each Y_t takes values in a common finite support $\mathcal{Y}_t$. Let $\rho_t := \rho\big(p_\theta^{(t)}, p_{\theta'}^{(t)}\big)$ be the Bhattacharyya affinity. Then the affinity between the product laws on horizon T is*

$$\rho\Big(\bigotimes_{t=0}^{T} p_\theta^{(t)}, \bigotimes_{t=0}^{T} p_{\theta'}^{(t)}\Big) = \prod_{t=0}^{T} \rho_t. \tag{S7.1}$$

*If there exists an infinite set $\mathcal{S}_{\theta,\theta'} \subseteq \mathbb{N}$ with $\|p_\theta^{(t)} - p_{\theta'}^{(t)}\|_{\mathrm{TV}} \geq \varepsilon > 0$ for all $t \in \mathcal{S}_{\theta,\theta'}$, then $\prod_{t\in\mathcal{S}} \rho_t = 0$ and, by the criterion of Kakutani (1948), the product measures over the infinite sequence are **mutually singular** (Shiryaev, 2019).*

Proof. **Identity** (S7.1)**.** Under independence,

$$\rho\Big(\bigotimes_{t=0}^{T} p_\theta^{(t)}, \bigotimes_{t=0}^{T} p_{\theta'}^{(t)}\Big) = \sum_{(y_0,\ldots,y_T)} \prod_{t=0}^{T} \sqrt{p_\theta^{(t)}(y_t) p_{\theta'}^{(t)}(y_t)} = \prod_{t=0}^{T} \sum_{y_t\in\mathcal{Y}_t} \sqrt{p_\theta^{(t)}(y_t) p_{\theta'}^{(t)}(y_t)}.$$

Singularity. The proof uses the standard inequality $\|p-q\|_{\mathrm{TV}} \leq \sqrt{2}\,\mathsf{H}(p,q)$ on finite spaces,

with $\mathsf{H}^2(p,q) = 1 - \rho(p,q)$. If $\|p - q\|_{\mathrm{TV}} \geq \underline{\varepsilon}$, then

$$\rho(p,q) \leq 1 - \frac{\underline{\varepsilon}^2}{2}.$$

At each $t \in \mathcal{S}_{\theta,\theta'}$, $\rho_t \leq 1 - \underline{\varepsilon}^2/2 < 1$. As $\mathcal{S}$ is infinite, the product contains infinitely many factors uniformly less than 1: $\prod_{t\in\mathcal{S}} \rho_t = 0$. By the criterion of Kakutani (1948), the infinite product measures are mutually singular (Shiryaev, 2019). □

Theorem S7.2 (Posterior consistency under product-measure singularity)**.** *Suppose Θ_K finite, prior μ_0 with full support, and PBE-consistent Bayesian updating (Definition S1.3). Let θ_K^* be the realized type and $\mathcal{T}$ the prolonged captivity event (Definition S3.1). Suppose that for all $\theta \neq \theta_K^*$ the following hold:*

(i) ***Trajectory compatibility.*** *$\mathbb{P}_{\mathrm{C},\theta_K^*}(\mathcal{T}) > 0$; for each false type, either $\mathbb{P}_{\mathrm{C},\theta}(\mathcal{T}) = 0$, or signal laws are compared under the law restricted to $\mathcal{T}$.*

(ii) ***Product structure and recurrent TV separation.*** *The sequence $\{Y_t\}_{t\geq 0}$ satisfies product factorization and the existence of $\mathcal{S}_{\theta,\theta_K^*}$ and $\underline{\varepsilon}_\theta > 0$ from Lemma S7.1.*

(iii) ***Signal-experiment alignment.*** *In $\mathcal{T}$, ζ_t contains $Y_t = (m_t, d_t)$, so that singularity of $\{Y_t\}$ transfers to $\{\zeta_t\}$ by monotonicity of singularity with respect to σ-algebras.*

Then, in $\mathcal{T}$,

$$\mu_t(\theta) \xrightarrow{t\to\infty} 0 \ \ \forall\, \theta \neq \theta_K^*, \qquad \mu_t(\theta_K^*) \xrightarrow{t\to\infty} 1, \qquad \mathbb{P}_{\mathrm{C},\theta_K^*}\text{-a.s.}$$

Equivalently, $\{\mu_t\}$ converges weakly to $\delta_{\theta_K^*}$ $\mathbb{P}_{\mathrm{C},\theta_K^*}$-a.s. in $\mathcal{T}$.

Proof. Fix $\theta \neq \theta_K^*$.

Case $\mathbb{P}_{\mathrm{C},\theta}(\mathcal{T}) = 0$: the false type cannot generate trajectories in $\mathcal{T}$; its posterior likelihood is annulled by survival exclusion.

Case $\mathbb{P}_{C,\theta}(\mathcal{T}) > 0$: by (ii), Lemma S7.1 implies that the infinite product laws are mutually singular. By (iii), that singularity transfers to the public flow $\{\zeta_t\}$. There exists a tail event A_θ with

$$\mathbb{P}_{C,\theta^*_K}(A_\theta \mid \mathcal{T}) = 1, \qquad \mathbb{P}_{C,\theta}(A_\theta \mid \mathcal{T}) = 0.$$

The likelihood ratio R^θ_t of the false type against the true type satisfies $R^\theta_t \to 0$ $\mathbb{P}_{C,\theta^*_K}$-a.s. in $\mathcal{T}$ (Blackwell and Dubins, 1962; Schervish, 1995). By the Bayesian updating rule defined in Definition S1.3,

$$\mu_t(\theta) \leq \frac{\mu_0(\theta)}{\mu_0(\theta^*_K)} R^\theta_t \to 0 \qquad \mathbb{P}_{C,\theta^*_K}\text{-a.s. in } \mathcal{T}.$$

As Θ_K is finite, all false coordinates converge to zero simultaneously. Finally,

$$1 = \lim_{t\to\infty} \sum_\theta \mu_t(\theta) = \lim_{t\to\infty} \mu_t(\theta^*_K) + 0,$$

so that $\mu_t(\theta^*_K) \to 1$ $\mathbb{P}_{C,\theta^*_K}$-a.s. in $\mathcal{T}$. □

S8 Formal verification in Lean 4

This replication package includes a standalone Lean 4 project in folder `lean/` (main file `AppendixBProofs.lean`) that formalizes the logical dependency graph of this supplement. The development is independent of earlier project-specific files and mirrors Sections S1–S7. Each declaration is tagged with its TeX label where applicable. A successful build (`lake build`) yields no `sorry` obligations. Build instructions appear in Section S9.

Reading the status tags. Every Lean declaration carries one of two tags. `[Proved]` means the statement is fully verified by the Lean 4 compiler against Mathlib: the proof is machine-checked and admits no gaps. `[Hypothesis]` means the statement is declared as an explicit interface and *assumed*, not re-proved, inside Lean; each such interface corresponds to a classical step proved analytically in this supplement or in the cited literature. The formalization therefore does *not* replace the proofs in this document: it certifies the finite-

dimensional algebra and the logical assembly of the identification argument, and makes every classical input explicit.

Proved in Lean (machine-checked).

- **Section S1.** Public filtration, predictive likelihood, and martingale beliefs; Lemma S1.4 (bounded martingale with a.s. limit in $[0,1]$, via Doob's theorem in Mathlib); and the normalization closure of step 3 of Theorem S3.6: false-type collapse forces $\mu_t(\theta^*) \to 1$.

- **Section S2.** Continuation probability of Definition S2.4; Cox hazard-ratio algebra (distinct $\beta_{K,j} \Rightarrow$ ratio $\neq 1$); the predictive masses $p_\theta^{(t)}$ form probability masses; local TV separation — Lemma S3.4(i) — for those masses under $\beta_{K,j}$ separation and a non-knife-edge total hazard; logit axioms (A1) full support and (A2) plan dominance of Remark S2.3.

- **Section S4.** Proposition S4.1: compact and signed forms of κ_h (`kappa_h_signed`, `dLambda_eq_kappa`); continuation factorization and (S4.5) (`dContProb_eq`); sign certificate of (S4.3) with explicit Δt and survival tail (`prop_expected_captivity_gamma`, `prop_expected_captivity_sgn`). Wald identity and DCT remain analytic.

- **Section S3.** Logical assembly of Lemma S3.4: conditions (a)/(b) imply Definition S3.2; packaging of recurrent uniform TV margins; and the Bayes factorization of step 2 of Theorem S3.6: the bound $\mu_t(\theta) \le (\mu_0(\theta)/\mu_0(\theta^*))\, R_t^\theta$ forces $\mu_t(\theta) \to 0$ whenever $R_t^\theta \to 0$.

- **Section S7.** Bhattacharyya product identity (S7.1); the Hellinger–TV bound $\rho \le 1 - \underline{\varepsilon}^2/2$; and vanishing of the product affinity under recurrent separation, i.e., the entire algebraic part of Lemma S7.1.

- **Identification bridge.** Cox predictive masses plus a recurrent TV margin imply asymptotic identification, conditional only on the two Kakutani interfaces listed below.

Not proved in Lean (explicit hypotheses). The following classical steps are stated as interfaces and assumed inside Lean, mainly because infinite product measures and their convergence theory are not available in Mathlib. Each one is proved analytically where indicated.

- *Kakutani mutual singularity* — vanishing product affinity implies mutually singular trajectory laws: proof of Lemma S7.1 and Kakutani (1948).
- *Singularity transfer* $Y_t \Rightarrow \zeta_t$: item (iii) of Theorem S7.2.
- *Singularity implies vanishing likelihood ratio*: step 2 of Theorem S3.6 (Blackwell and Dubins, 1962; Schervish, 1995).
- *Survival exclusion*, $\mathbb{P}_{C,\theta}(\mathcal{T}) = 0$ implies collapse: step 2 of Theorem S3.6.
- *Maximum entropy axiom (A3)* of Remark S2.3: Jaynes (1957).
- *Implicit-function step* of Proposition S5.1: proof in Section S5.
- *Existence of the implementable PBE*, Theorem S6.6: fixed-point proof in Section S6 (Kakutani, 1941).

Table S8.1 maps these hypotheses to their Lean identifiers.

S9 Reproducibility

All analytical proofs in this supplement are self-contained in `Appendix_B_Proofs.tex`. Machine-checked algebraic steps are reproduced by the Lean 4 project in `lean/`, openly available in the public repository at https://github.com/Donzhumber/TPV.git. Set the working directory to `appendix_B_proofs/` and run `replicate.sh`. See `README.txt` at the root of that folder for software requirements (Lean 4, elan, `lean-toolchain`) and Lake; Mathlib is fetched automatically on first build (network required; estimated first-build time 15–45 minutes).

Table S8.1: Lean formalization status for selected supplement objects

TeX object	Lean identifier	Status
Lemma S7.1 (algebraic part)	`generalIdAppendixLemma`	Proved
Kakutani (1948) mutual singularity	`kakutaniMutualSingularity`	Hyp.
Outcome-trajectory laws on $\mathcal{T}$	`kakutaniSignalSingularity`	Hyp.
Singularity transfer $Y_t \Rightarrow \zeta_t$	`signalExperimentAlignment`	Hyp.
Singularity $\Rightarrow$ vanishing likelihood ratio	`singularityImpliesLikelihoodVanishing`	Hyp.
$\mathbb{P}_{C,\theta}(\mathcal{T}) = 0 \Rightarrow$ collapse	`zeroMassPosteriorCollapse`	Hyp.
Remark S2.3, axiom (A3)	`logitProb_maxEntropy_hyp`	Hyp.
Proposition S4.1, eq. (S4.2) (signed form)	`kappa_h_signed`, `dLambda_eq_kappa`	Proved
Proposition S4.1, eq. (S4.5)	`dContProb_eq`	Proved
Proposition S4.1, eq. (S4.3) (sign)	`prop_expected_captivity_gamma`, `prop_expected_captivity_sgn`	Part.
Proposition S5.1	`prop_optimal_gamma_beliefs`	Hyp.
Theorem S6.6	`conditionalImplementability`	Hyp.

Note. `[Proved]`: machine-checked in Lean 4. `[Part.]`: partial certificate (sign of (S4.3) given the analytic chain). `[Hyp.]`: declared interface; proved analytically in this supplement or cited literature. *Source:* Analytical proofs in this supplement; Lean identifiers in `lean/AppendixBProofs.lean`. Software: Lean 4 (`lean-toolchain`); Mathlib via Lake.

1. `replicate.sh` at the package root runs `lake build` in `lean/` and writes a log to `lean/build.log`.

2. Main formal file: `lean/AppendixBProofs.lean`. Project metadata: `lakefile.lean`, `lean-toolchain`, `lake-manifest.json`.

3. Section S8 and Table S8.1 document which results are `[Proved]` vs. `[Hyp.]`.

4. (Optional) Compile `tex/Appendix_B_Proofs.pdf` from `tex/` with `pdflatex` and `bibtex`.

Section S8 (Table S8.1) documents which results are machine-checked vs. hypothetical. All paths are relative to `Appendix_B_Proofs/`; see `README.txt` for the full build workflow.

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